Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*		[It has] magnitude (accept size) and direction	1			1		
	(b)	(i)	Total momentum before collision = 30 000 + 15 000 [= 45 000 kg m s^{-1}] (1) Total momentum after collision = 27 000 + 18 000 [= 45 000 kg m s^{-1}] (1) Ignore units Deduct 1 mark for powers of 10 slip Award 1 mark only for - Momentum is not lost [in collision] or momentum before [collision] is the same as momentum after [collision] / momentum is conserved (1) Don't accept: they are the same **Alternative:** Loss in momentum of A =12 000 [kg m s^{-1}] (1) Gain in momentum of B = 12 000 [kg m s^{-1}] (1) Hence the gain in momentum of B = loss in momentum of A (1)	1	1 1		3	2	
		(ii)	Attempt at using $p_A + p_B = (m_A + m_B)v$ (1) Correct substitution e.g. 45 000 **ecf** = 25 000v (1) Award 2 marks if this seen. v = 1.8 [m s^{-1}] (1)	1	1 1		3	3	
	(c)	(i)	A body's <u>rate of</u> / change per second (reference to time) change of momentum (1) is proportional to [accept 'equal to'] the [resultant] force acting on it (1) [and is in the direction of this force] **Alternative:** Formula stated (1) with all terms defined (1)	2			2		
		(ii)	Time for collision = 0.2 s (1) accept (0.5 – 0.3) $F = \frac{-12\,000}{0.2}$ (1) F = -60 000 [N] (1) **ecf** on powers of 10 slip		1 1	1	3	3	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Newton's 3rd Law (1) Accept N3 Law Change of momentum is +12 000 kg m s^{-1} or the same **and** collision time is the same (1) Accept: [magnitude] of gradient same Don't accept graph is symmetrical	1	1		2		
			Question 7 total	**6**	**7**	**1**	**14**	**8**	**0**